MIL-B-36382 (DM)
10 February 1965

MILITARY SPECIFICATION

BATH, CONSTANT TEMPERATURE

1. SCOPE

1.1 This specification covers one type of constant temperature oil bath, designed to provide an extremely stable and uniform temperature environment for standard cells, conductivity cells, resistance standards, and similar laboratory devices.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

STANDARD

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

FSC 6640

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3.2 Construction. The constant temperature bath shall consist essentially of a working tank, stirrer, immersion heaters, and oil flow detector, all enclosed in an insulated oil recirculation tank; a temperature control system; a cooling system; and a control console. The unit shall be housed in a metal cabinet, divided into three compartments - upper, lower, and right side. The control console shall be mounted atop the cabinet.

3.2.1 Oil recirculation tank. The oil recirculation tank shall be enclosed in the upper compartment of the cabinet. The sides of the recirculation tank shall extend to the top of the cabinet. A folding cover, hinged to the top surface of the cabinet, shall form the top of the recirculation tank and shall permit ready access to the interior. The recirculation tank shall be constructed of stainless steel and shall have a capacity of approximately 36 gallons. The recirculation tank shall support an inner working tank and shall also contain an oil flow detector, a motor-driven stirrer, and two immersion heaters. The oil flow detector shall be located beneath the inner working tank. The stirrer and immersion heaters shall be in a compartment alongside the inner working tank. There shall be sufficient clearance along the sides and bottom of the inner working tank to allow free overflow and circulation of oil. The recirculation tank shall be so designed that the stirrer circulates the oil through the oil flow detector upwards through and overflowing the inner working tank, and then past the heaters back to the stirrer. The recirculation tank shall be insulated with a 1-1/2-inch layer of a cellular sponge on all sides and on the bottom.

3.2.1.1 Working tank. The working tank shall be constructed of stainless steel and shall be enclosed in the oil recirculation tank. Nominal inside dimensions of the working tank shall be 18 inches long, 18 inches wide, and 13-1/2 inches deep. The working tank shall be provided with a perforated stainless steel platform, which can be suspended at either 7-1/2 inches or 9 inches depth. Openings in the bottom of the working tank shall permit circulation of oil through the tank. The working tank shall accommodate a maximum of 60 standard saturated cells. There shall be sufficient clearance along the sides and bottom of the working tank to permit free circulation of oil.

3.2.1.2 Immersion heaters. Two 650-watt nichrome electric heaters shall be provided. The heaters shall be mounted in the upper section of the compartment, which is alongside the inner working tank. One heater shall be continuously controlled by the temperature control system. The other

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heater shall be connected to line voltage and shall be controlled by a Variac and "ON-OFF" switch. This heater shall be used to provide auxiliary heat for quick start-up or high-temperature operation.

3.2.1.3 Oil flow detector. The oil flow detector shall consist of two nickel-coil resistance thermometers, concentrically mounted below the inner working tank.

3.2.1.4 Stirrer. The stirrer shall be constructed of stainless steel and shall consist of a stirring rod and blade. The stirrer shall be suspended vertically from the top of the oil recirculation tank, and shall be located in the compartment alongside the inner working tank. The stirrer blade shall be positioned below the two immersion heaters. The stirrer shall be powered by a constant-speed motor, located just outside the oil recirculation tank in the right side compartment of the metal cabinet. The stirrer shall adequately circulate a suitable oil through the system described in 3.2.1.

3.2.1.5 Cover. The recirculation tank shall be covered by a folding cover which forms part of the top surface of the cabinet. The cover shall consist of two glass panels, enclosed in stainless steel frames. The cover shall be hinged to the top of the cabinet and shall permit ready access to the interior of the tank.

3.2.2 Temperature control system. The temperature control system shall consist of the following components:

(a) A resistance thermometer bridge containing two 1000-ohm nickel-coil thermometers, a fixed resistor, and three decades and a slide wire for control point setting.

(b) An electronic, direct-current, linear, null detector.

(c) A current-adjusting control unit.

(d) A silicon-controlled rectifier unit.

(e) Immersion heaters.

The null detector, current-adjusting control unit, and the rectifier unit shall be located in the lower compartment of the cabinet. The immersion heaters and nickel-coil thermometers shall be in the oil recirculation tank as described in 3.2.1.2 and 3.2.1.3. The fixed resistor and the three decades and slide wire shall be positioned just outside the oil recirculation tank in the upper compartment of the cabinet. The bridge control dials shall be located atop the metal cabinet. The temperature control system shall operate as follows : When the bath is set for a desired temperature,a bridge unbalance

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is developed. The bridge unbalance is sensed by the null detector, and the null detector output is applied to the current-adjusting control unit. The control unit provides an output to the rectifier which regulates power to the immersion heaters.

3.2.3 Cooling system. The cooling system shall consist of copper tubing coils which surround the oil recirculation tank and a pressure regulator and needle valve to control the flow of coolant. The needle valve and pressure regulator shall be located in the right side compartment of the cabinet, and shall be provided with adequate fittings for connection to the coolant supply.

3.2.4 Control console. The control console shall be located atop the cabinet and shall consist of the following components:

(a) Overheat control thermostat with associated pilot light and reset button.

(b) Ammeter to indicate current through the controlled heater.

(c) High control switch for controlled heater.

(d) Control "ON-OFF" switch for energizing control system.

(e) Auxiliary meter for null detector.

(f) Line "ON-OFF" switch and associated pilot light.

(g) Variac for energizing auxiliary heater and associated "ON-OFF" switch.

(h) Switch for energizing bridge circuit and associated pilot light.

3.2.5 Cabinet. The bath unit shall be housed in a metal cabinet which is divided into an upper, lower, and right side compartments. The upper compartment shall contain the oil recirculation tank, cooling coils, fixed resistor, and decade box. The lower compartment shall contain the null detector, current-adjusting control unit, and silicon carbide rectifier, and shall be of the sliding-drawer type to permit ready access. The right side compartment shall contain the stirrer motor, the coolant pressure regulator, and the coolant needle valve, and shall permit ready access to the interior of the bath. The control console shall be mounted atop the cabinet. Nominal overall dimensions including the control console shall be 50-3/4 inches high, 43-1/8 inches wide, and 26-1/8 inches deep. The nominal height to work surface (cover of recirculation tank) shall be 40-3/4 inches. An eight-watt fluorescent light and associated "ON-OFF" switch shall be provided to illuminate the interior of the tank.

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3.3 Performance.

3.3.1 Operating range. The bath shall have an operating range of 0° centigrade (C.) to 100° C.

3.3.2 Withstand voltage. The bath shall be capable of withstanding successfully for 1 minute the application of 1000 volts, 60 cycles, when applied between current-carrying parts and metallic non-current-carrying parts with the bath at the maximum operating temperature attained in normal use, when tested as specified in 4.4.1.

3.3.3 Constancy and uniformity. The bath temperature shall be maintained to within ± 0.001° C. of the set point in the range of 15° C. to 60° C. and to within ± 0.003° C. of the set point in the range of 0° C. to 15° C. and 60° C. to 100° C., when tested as specified in 4.4.2.

3.3.4 Temperature control setting. The temperature control setting shall be adjustable in steps of less than 0.001° C.

3.3.5 Power supply. The constant temperature bath shall be equipped to operate on 120 volts, 50 or 60 cycles, alternating current.

3.4 Identification marking. Each principal component or assembly shall be permanently and legibly marked with the name or registered trademark and model number of the manufacturer.

3.5 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. In addition, the contractor shall mail 15 copies of the service data to the contracting officer for distribution.

3.6 Workmanship. Workmanship shall be first class throughout. The bath unit shall be free from defects which detract from its appearance or may impair its serviceability. All metal surfaces shall be smoothly finished.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the bath.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Withstand voltage	3.3.2	4.4.1	100%	*
Constancy and uniformity	3.3.3	4.4.2	100%	*

* Acceptance number shall be zero.

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4.3 Examination. The baths shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Component parts missing, cracked, or broken.
102 Cover does not open and close freely.
103 Cover does not rest tightly in place when closed.
104 Component parts improperly positioned in cabinet.
105 Lower compartment of cabinet does not slide in and out freely.

Minor

201 Metal edges and surfaces not smoothly finished and free of sharp edges.
202 Right panel compartment does not open and close freely.
203 Identification marking not complete, correct, legible, or permanent.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The baths shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Withstand voltage test. Voltage shall be applied to the bath using a testing transformer with adjustable output voltage. Starting at zero volts, the applied voltage shall be increased gradually until the required voltage is reached.

4.4.2 Constancy and uniformity. The bath temperature shall be determined by a platinum resistance thermometer connected to a Mueller bridge. The platinum resistance thermometer shall be a secondary standard thermometer which has been checked by the National Bureau of Standards. The test shall be run with the maximum number of standard cells in the bath, the cover in place, and with oil circulating through the bath as in

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normal operating conditions. Temperature determinations shall be made at 10° C. intervals throughout the entire range of the bath.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Level C. The commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Interior packages. Each package shall be marked to identify contents.

5.2.2 Exterior containers. Exterior containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and contract or purchase order number. The legend "OPEN THIS SIDE" shall be shown, where applicable, to facilitate locating service data and inspecting and removing the unit.

6. NOTES

6.1 Ordering data. Procurement documents should specify the number and date of this specification.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-1H004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

BATH, CONSTANT TEMPERATURE, MIL-B-36382, (DM) 10 February 1965

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE DMS)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

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